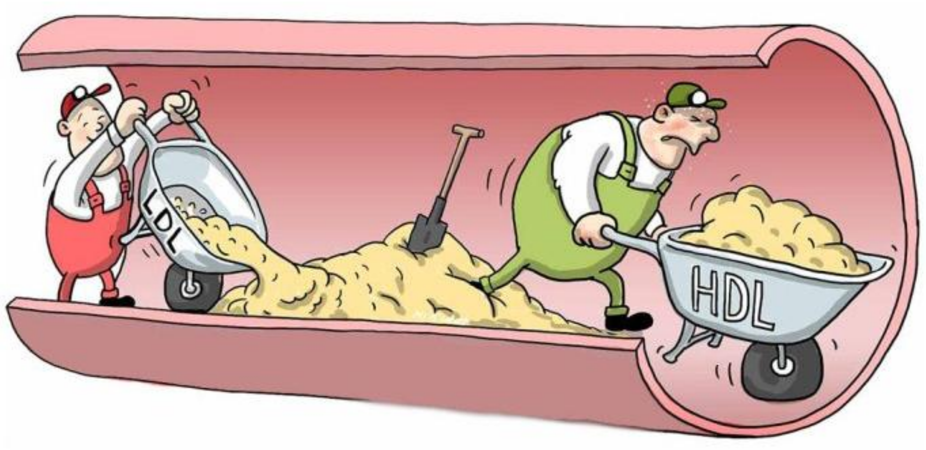
Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 16
HDL Metabolism and Lipoprotein Disorders



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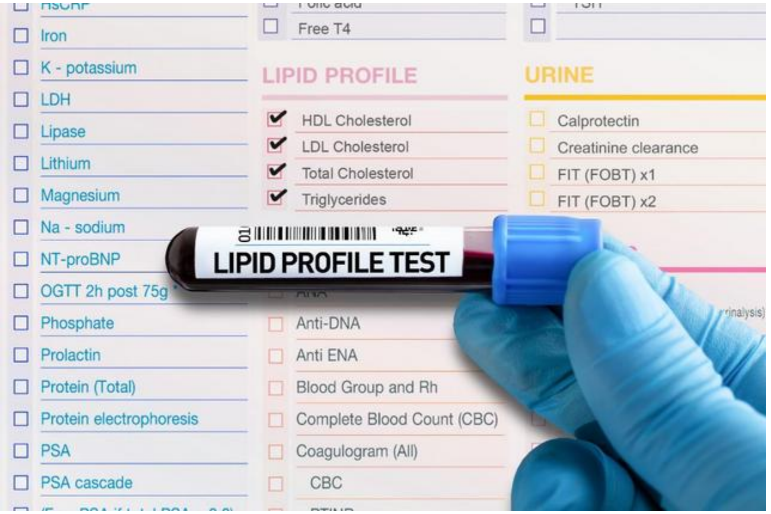
Lecture: HDL and Lipoprotein Disorders

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0223	Discuss the functions and effects of deficiency of apo B-48, apo B-100, apo C-II, apo E.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0225	Describe reverse cholesterol transport and the role of HDL and SRB-1 receptors.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0226	Describe the synthesis, location and action of lecithin: cholesterol acyltransferase (LCAT) and activation by apo A-1.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0227	Describe the function of cholesterol ester transfer protein (CETP) and explain its role in cholesterol ester transfer from HDL to VLDL, IDL and LDL.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0229	Describe hypoalphalipoproteinemia (Tangier disease) and abetalipoproteinemia and hypobetalipoproteinemia (molecular mechanism, lipid profile abnormalities, basis for clinical features).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0231	Differentiate different hyperlipidemias (Type I, IIa, IIb, III, IV and V) using lipid profile (Serum TAG and Serum cholesterol), lipoprotein electrophoresis, clinical features and molecular mechanisms and inheritance patterns.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0232	Review the other names for the hyperlipidemias type I-V.

Recommended Reading

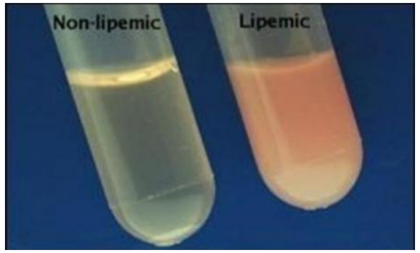
- Lippincott's Biochemistry 9th Edition 🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
  - Chapter 18: Cholesterol, lipoprotein and steroid metabolism; Section: Lipoproteins
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition; Chapter 18; Section: Lipoproteins; Cholesterol, Lipoprotein, and Steroid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library **Study questions: 18.2; 18.3; 18.4**
- Mark's Basic Medical Biochemistry: A clinical approach **(MBMB**) 6th edition; Sections: V, VI, VII, VIII; Cholesterol Absorption, Synthesis, Metabolism, and Fate | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library
- Rapid review Pathology: Goljan https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323476683000107

Fasting Lipid profile estimation: Triglycerides



- Serum Triacylglycerol (Triglycerides; TG)
  - Chylomicrons and VLDL
  - Chylomicrons are normally **not present** in fasting state
    - If chylomicrons present, indicates abnormal chylomicron metabolism

- **Elevated triglycerides (TG)**



Lipemic plasma (milky, turbid)

Triglyceride ranges for adults

Normal	Less than 150 mg/dL
Borderline high	150-199 mg/dL
High	200-499 mg/dL
Very high	500 mg/dL or higher

Source: National Institutes of Health

Serum cholesterol and fractions

- Total cholesterol
  - Elevated Total cholesterol or LDL cholesterol: Plasma clear
- Serum Cholesterol fractions:
  - HDL cholesterol
  - **LDL cholesterol** (Direct estimation difficult)
  - Friedewald formula: For LDL cholesterol calculation

LDL cholesterol = Total cholesterol – HDL cholesterol – (TG/5)

Clear plasma

TG/5 = VLDL-cholesterol (20% of VLDL is cholesterol)

(Formula not reliable when TG is very high)

TG = 250mg/dL; Total cholesterol = 270mg/dL; HDL-cholesterol = 40mg/dL. Calculate LDL-cholesterol.

LDL-cholesterol = 270 – 40 – (250/5) = 270 – 40 – 50 = 180mg/dL

Dyslipidemia increases ASCVD risk

- Elevated total cholesterol
- Elevated LDL-cholesterol (Clear plasma)
- Low HDL-cholesterol
- Elevated triglycerides (TG) (Lipemic plasma)
- Elevated Triglyceride/ HDL ratio: >2
- Lipoprotein(a)
- LDL-B (Small dense LDL)

- Family history of cardiovascular disease
- Obesity, Smoking, Age, hypertension, Diabetes mellitus
- **Increases risk of ASCVD** (atherosclerotic cardiovascular disease)
https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimate/

Cholesterol Levels

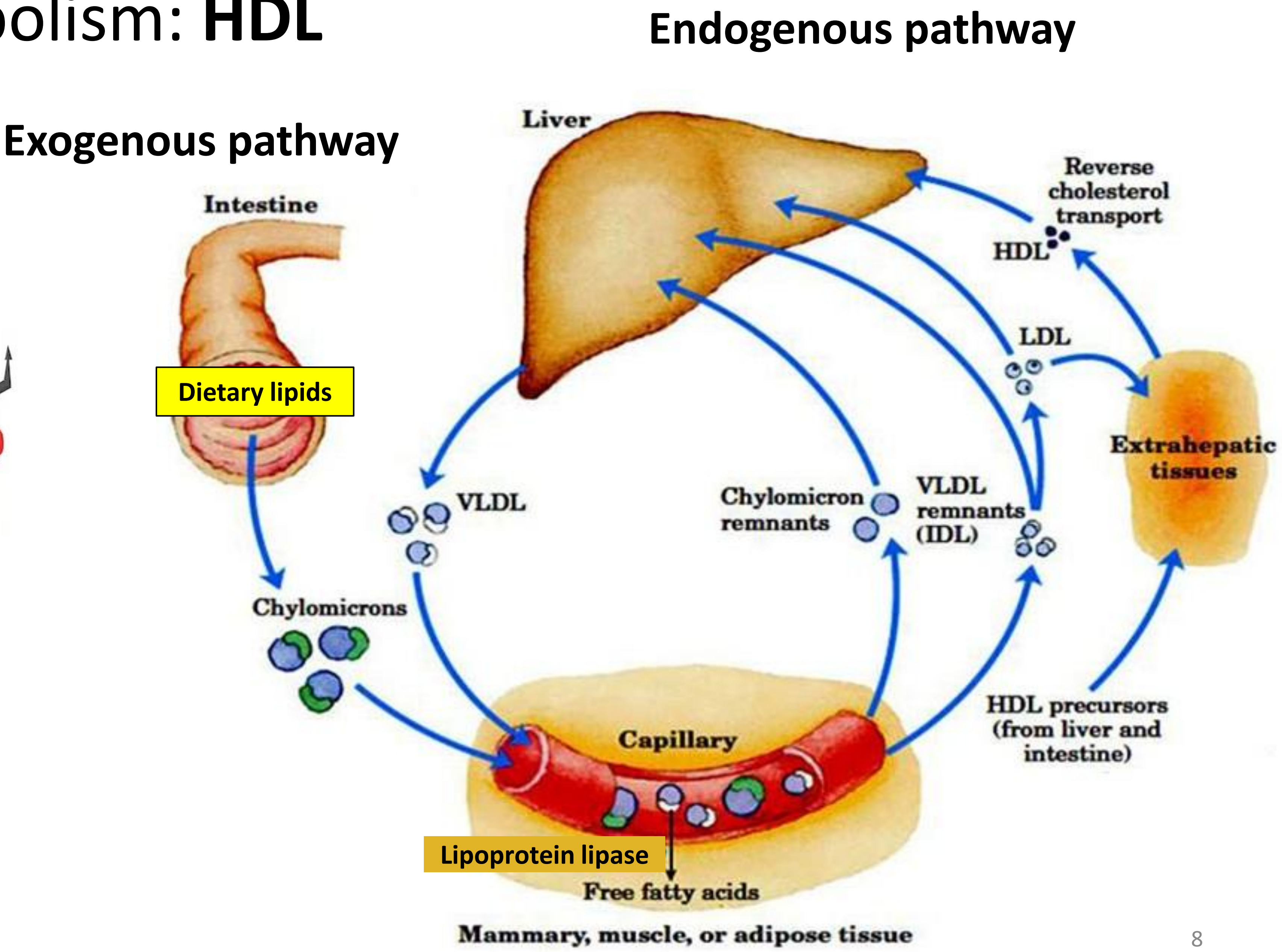
	Total Cholesterol	LDL Cholesterol	HDL Cholesterol
DANGEROUS	**240 and higher**	**160 and higher**	**Under 40** (male) **Under 50** (female)
AT-RISK	**200 – 239**	**100 – 159**	**40-59** (male) **50-59** (female)
HEART-HEALTHY	**Under 200**	**Under 100**	**60 and higher**

Triglyceride ranges for adults

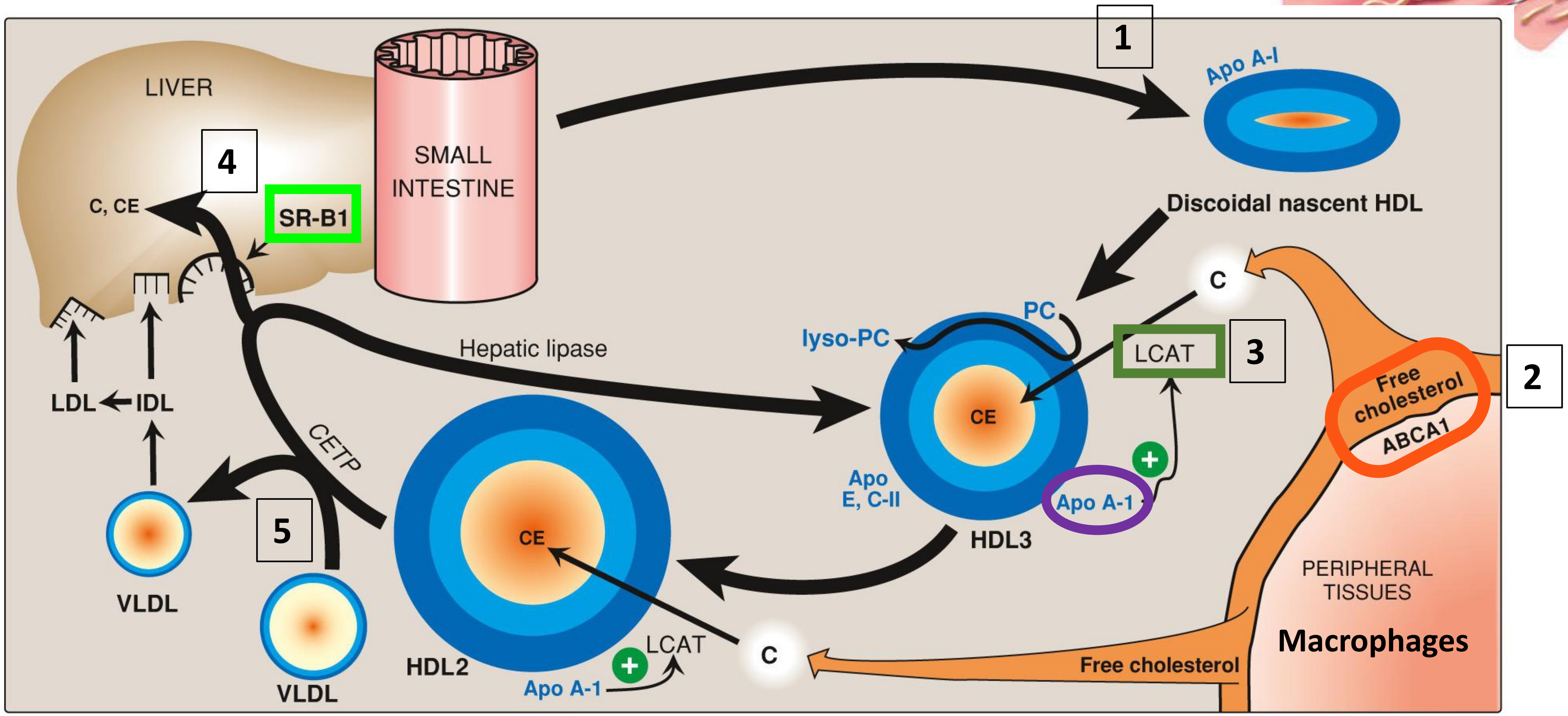
Normal	Less than 150 mg/dL
Borderline high	150-199 mg/dL
High	200-499 mg/dL
Very high	500 mg/dL or higher

Source: National Institutes of Health

Lipoprotein metabolism: **HDL**



HDL metabolism: Reverse cholesterol transport



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HDL metabolism: Reverse cholesterol transport

1. Liver and intestine form nascent HDL (discoidal; contains **Apo A-I**)
2. **ABCA1** in peripheral tissues and macrophages (foam cells) facilitate transport of free cholesterol out of the cells
   - Remember, cholesterol is a component of plasma membrane
3. **Lecithin cholesterol acyl transferase (LCAT)** converts cholesterol into cholesterol ester which enters the core of HDL particle
   - LCAT activated by ApoA-I
4. HDL is taken up by the liver via **SRB-1** (**Scavenger receptor-B1**)
5. Cholesterol ester transferred to VLDL by Cholesterol ester transfer protein (**CETP)**

Functions of HDL:

HDL transports cholesterol from peripheral tissues to the liver (reverse cholesterol transport); High HDL-cholesterol → Lower risk of ASCVD

HDL donates **ApoC-II and ApoE** to nascent VLDL and nascent chylomicrons;

HDL accepts **ApoC-II** from chylomicrons and VLDL which are then converted to remnant VLDL (IDL) and remnant chylomicrons

Role of ABCA1 cholesterol transporter

- Facilitates free cholesterol efflux from peripheral cells and macrophages
- Facilitates removal of cholesterol from tissues and foam cells
- **Tangier disease: Inherited ABCA1 cholesterol transporter deficiency** (hypoalphalipoproteinemia)
    - Cholesterol accumulates in tissues (**Orange tonsils**, corneal clouding)
    - Higher risk of early ASCVD and myocardial infarction
    - Very low HDL-cholesterol levels

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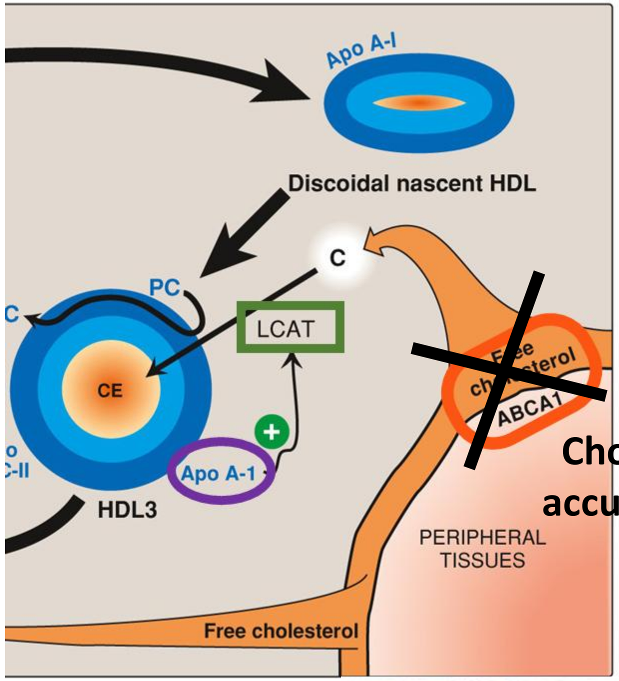
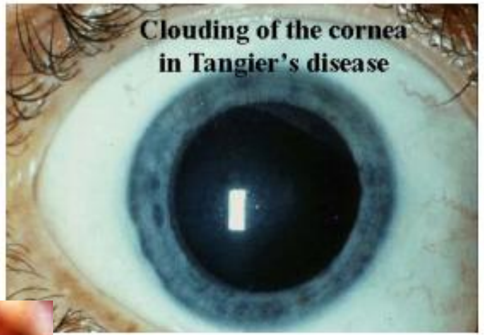


Cholesterol accumulation

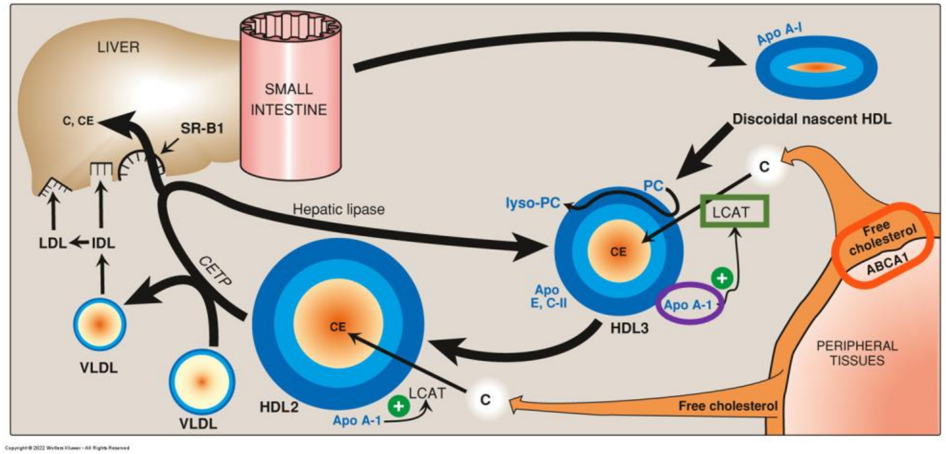
Corneal clouding



Orange tonsils

Lecithin cholesterol acyl transferase (LCAT)

- Catalyzes formation of cholesterol ester (hydrophobic/ non-polar)
  - LCAT is an enzyme in circulation
  - Differentiate from ACAT
- Cholesterol ester enters HDL core
- HDL taken up by SR-B1 (Scavenger receptor-B1) in liver
- Hence, LCAT helps in removal of cholesterol from cells and macrophages (reverse cholesterol transport)
- **LCAT deficiency:**
  - **Low HDL levels**
  - **And higher risk of ASCVD**



Free Cholesterol

Phosphatidylcholine (Lecithin)

Lecithin cholesterol acyl transferase (LCAT)

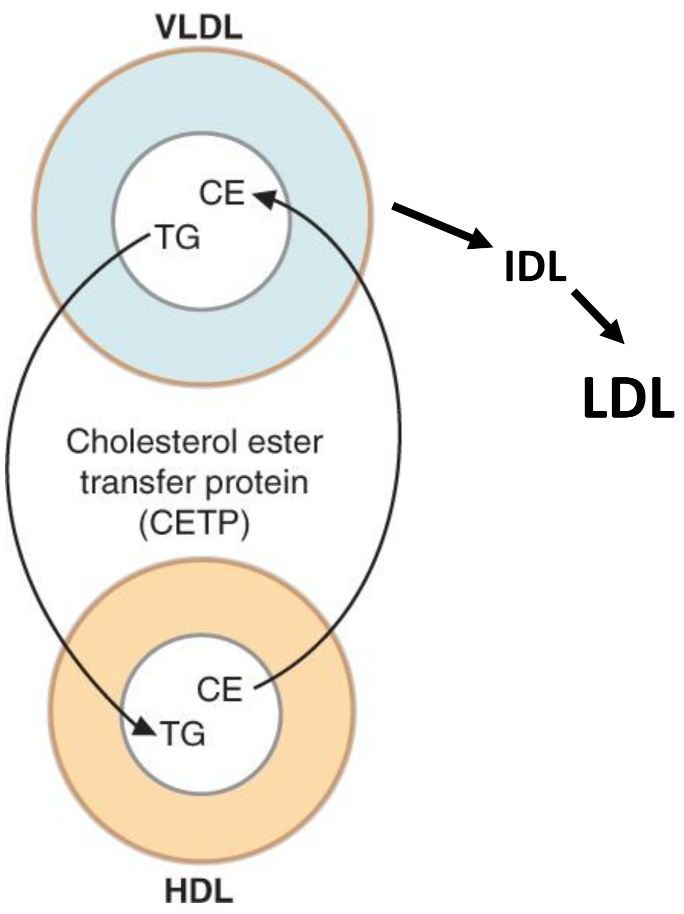
Apo A-I

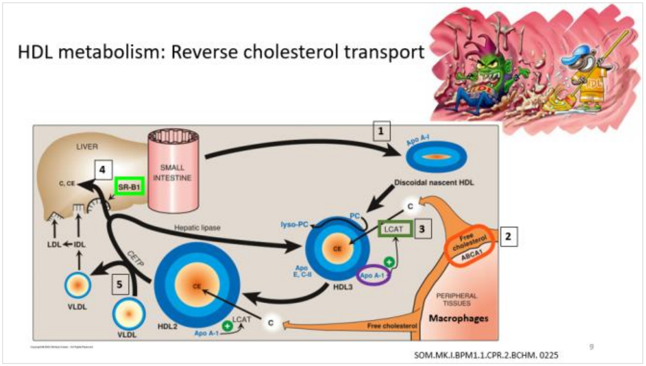
Cholesterol ester (Cholesterol + Fatty acid)

Cholesterol ester enters HDL core

Role of Cholesterol ester transfer protein (CETP)

- Transfers cholesterol ester (5) from HDL to VLDL → IDL → LDL
- **Higher CETP activity: Lowers HDL cholesterol and increases LDL cholesterol**
- **Lower CETP activity: Lowers LDL cholesterol and increases HDL cholesterol**





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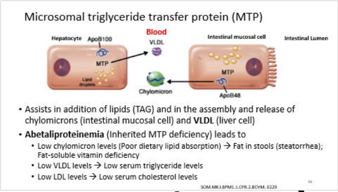
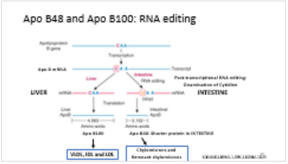
Hypolipoproteinemia/ Hypolipidemia

- Abetalipoproteinemia (MTP deficiency)



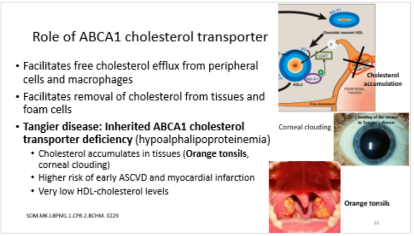
- Hypobetalipoproteinemia (Apo B gene mutations)

Disorders have similar clinical features and similar lab findings



  - **Low chylomicron (ApoB48)** → Impaired dietary fat absorption; Fat-soluble vitamin deficiency (vitamins A,D,E,K)
  - **Low VLDL, LDL (ApoB100)** → Low serum triglycerides (TAG) and low serum cholesterol

- Hypoalphalipoproteinemia (Low HDL-cholesterol): Increased risk of ASCVD
  - Inherited (Tangier disease): ABCA1 mutation
    - **Orange tonsils**, Increased risk of **ASCVD**
    - Low HDL-cholesterol levels
    - **Impaired reverse cholesterol transport**

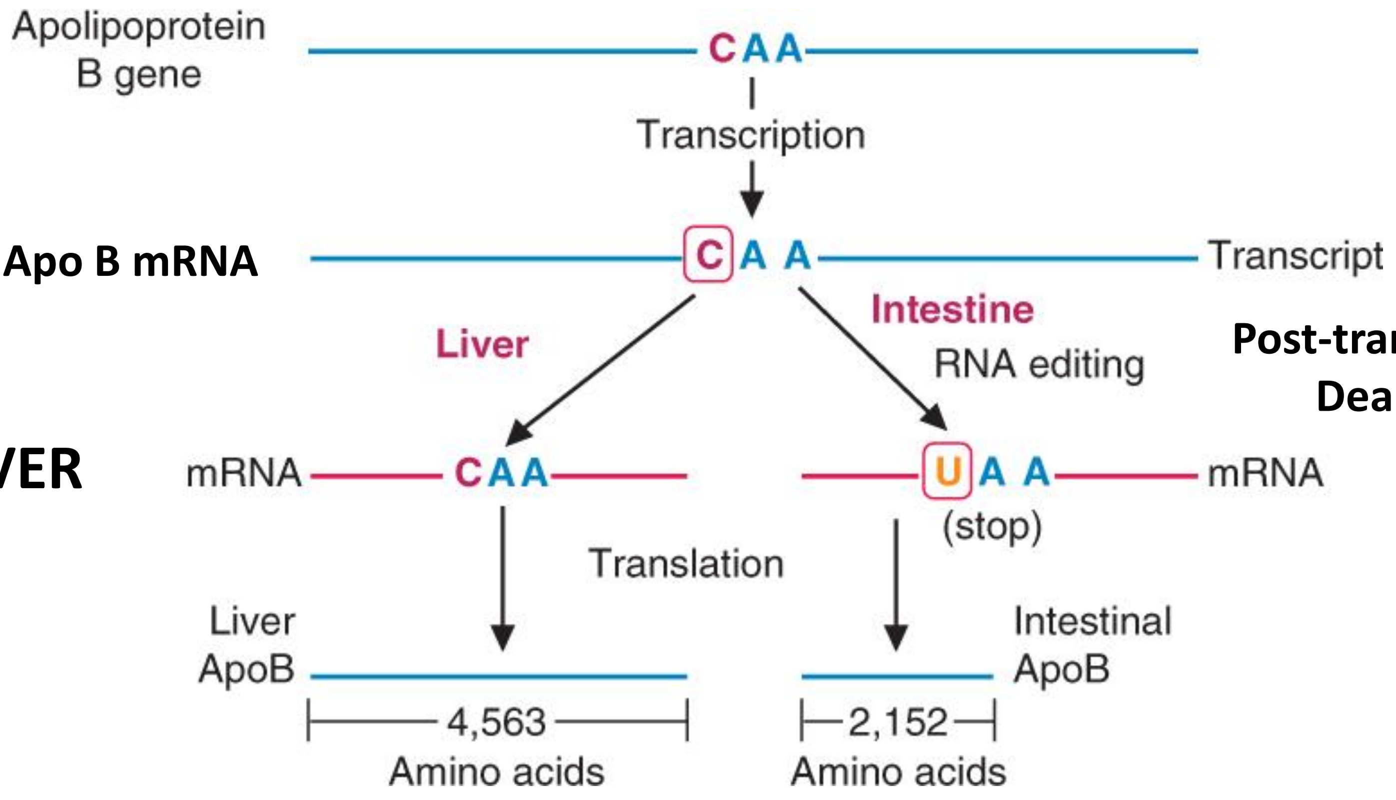


  - Acquired: Obesity, type 2 diabetes mellitus, Hypertriglyceridemia, smoking
    - **Low** HDL-cholesterol → Increased risk of **ASCVD**

Apo B48 and Apo B100: RNA editing

Apo B mRNA

LIVER



Post-transcriptional RNA editing: Deamination of Cytidine

INTESTINE

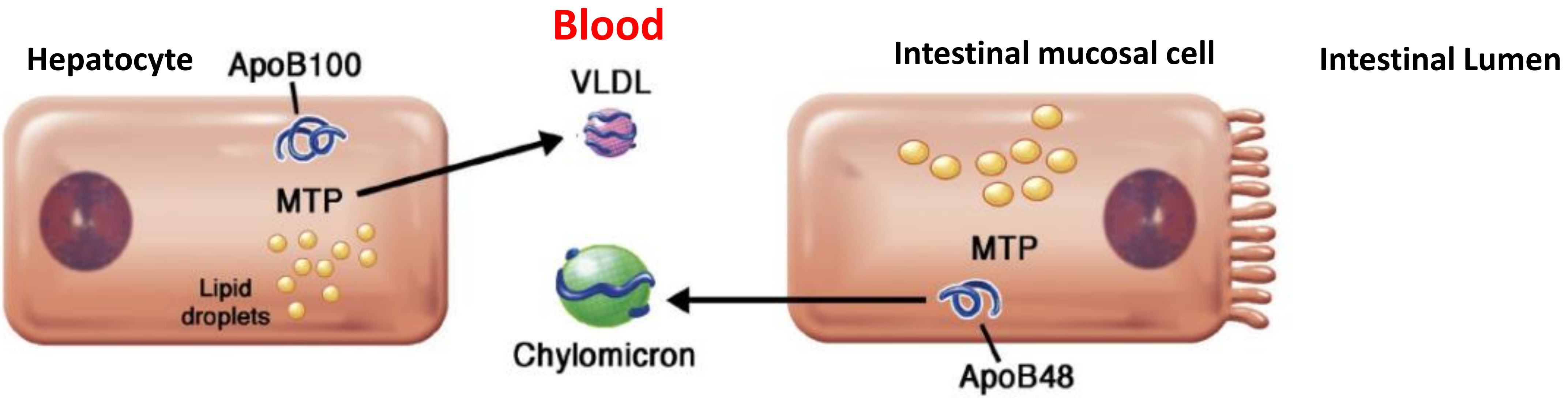
Apo B100

Apo B48: Shorter protein in INTESTINE

VLDL, IDL and LDL

Chylomicrons and Remnant chylomicrons

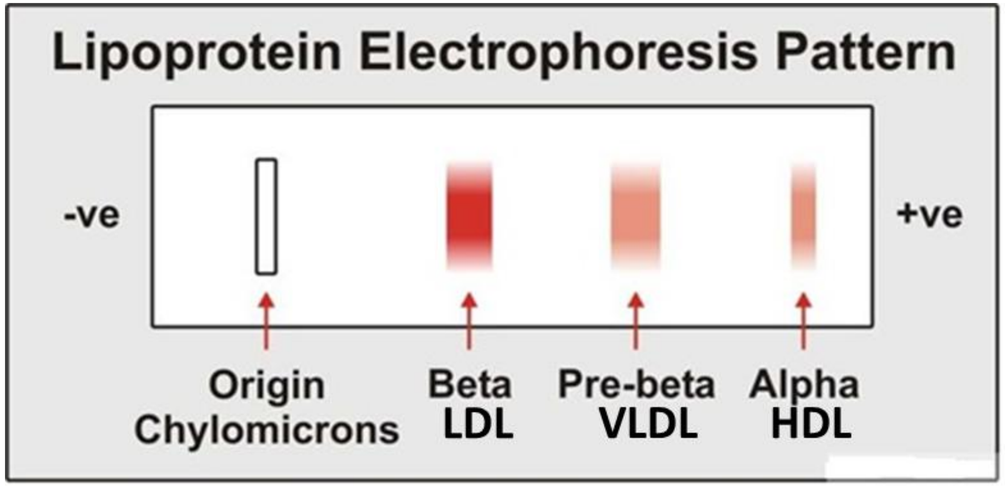
Microsomal triglyceride transfer protein (MTP)



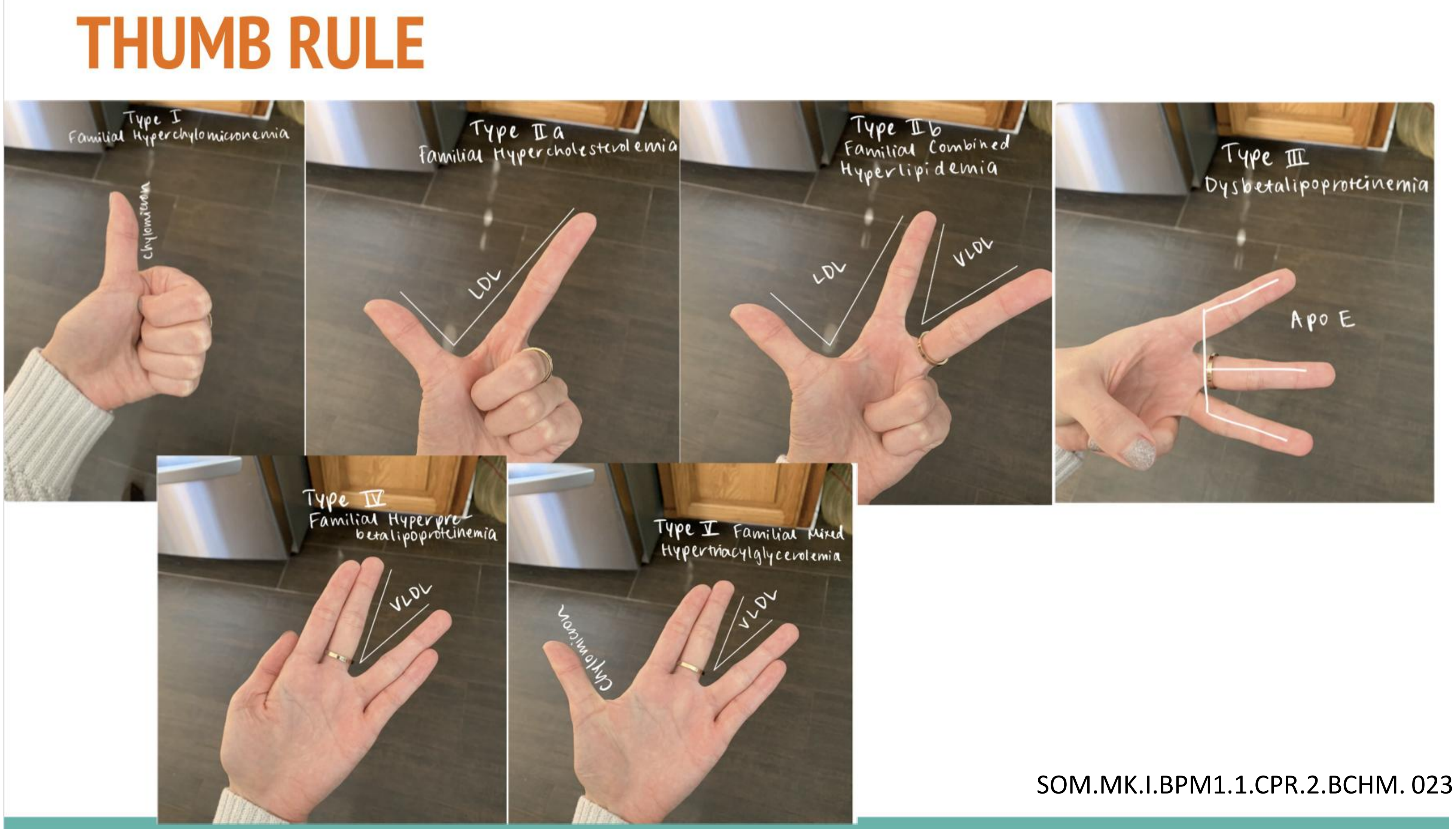
- Assists in addition of lipids (TAG) and in the assembly and release of chylomicrons (intestinal mucosal cell) and **VLDL** (liver cell)
- **Abetaliproteinemia** (Inherited MTP deficiency) leads to
  - Low chylomicron levels (Poor dietary lipid absorption) → Fat in stools (steatorrhea); Fat-soluble vitamin deficiency
  - Low VLDL levels → Low serum triglyceride levels
  - Low LDL levels → Low serum cholesterol levels

Hyperlipoproteinemias (Hyperlipidemia)

- Frederickson's phenotypes: Based on specific lipoprotein elevated (Electrophoresis or Ultracentrifugation)
  - Type I: Elevated chylomicrons
  - Type IIA: Elevated LDL (β lipoprotein)
  - Type IIB: Elevated LDL and VLDL (Pre-β lipoprotein and β lipoprotein)
  - Type III: Elevated IDL (remnant VLDL)
  - Type IV: Elevated VLDL (Pre-β lipoprotein)
  - Type V: Elevated Chylomicrons and VLDL



Provided by a student: Use it if it helps!!



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Hyperlipidemias Mnemonic: Use it if it helps!! **Or make your own!!**

1LP, 2LD, IIB add V, 3 is E, 4 gets more, (5 = Types I+IV)

Type I: 1LP: LPL deficiency (or Apo C-II) → ↑ chylomicron ,↑ TAG and pancreatitis

IIA LD: LDL receptor deficiency (or Apo B100) → ↑ LDL cholesterol, Achilles tendon xanthoma, corneal arcus (IIA)

IIB add V: ↑ LDL cholesterol add VLDL in type (IIB)

Type III: 3 is E: Apo E deficiency → ↓ remnant uptake, ↑ IDL, ↑ chylomicron remnants, palmar xanthoma

4 gets more: → ↑ VLDL due to over production; ↑ ↑ TAG; pancreatitis

Hyperlipoproteinemias (Hyperlipidemia)

- Based on the **fasting lipid profile** (Link to Frederickson phenotype)
  - **Hypertriglyceridemia**
    - Type I
    - Type IV
    - Type V
  - **Hypercholesterolemia**
    - Type IIA
  - **Hypertriglyceridemia and hypercholesterolemia**
    - Type IIB
    - Type III
- Inherited vs Acquired hyperlipidemia
  - Inherited: Due to single gene defects
  - **Acquired/ Secondary:** More common; Usually multifactorial (Effect of multiple genes and environmental factors)

Secondary (Acquired) hyperlipidemia

- Obesity, insulin resistance syndrome (metabolic syndrome), Type-2 diabetes mellitus
  - Can result in **type IV** (elevated VLDL) or **type IIB** (elevated LDL and VLDL)
  - **Low** activity of **lipoprotein lipase** (insulin stimulates LPL activity)
  - Also associated with **High LDL and low HDL** (Atherogenic)
  - Increases risk of **ASCVD**
- Hypothyroidism
  - Increases LDL-cholesterol; **Fewer LDL-receptors** on cell membrane
- Nephrotic syndrome
  - Increases LDL-cholesterol
- Alcohol use disorder
- Smoking …..

Primary (Inherited) hyperlipidemias

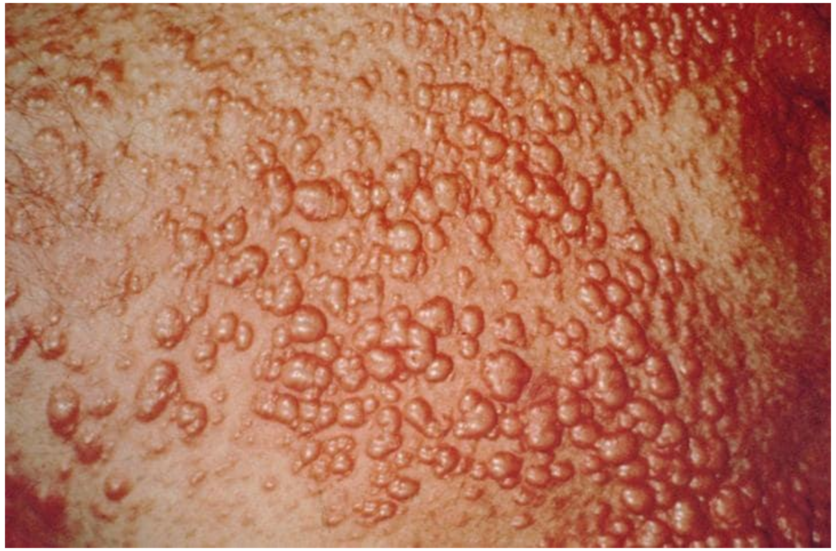
- Type I: Elevated chylomicrons **(Familial hyperchylomicronemia)**
- Type IIA: Elevated LDL (β lipoprotein) **(Familial hypercholesterolemia)**
- Type IIB: Elevated LDL and VLDL (Pre-β and β lipoprotein) **(Familial combined hyperlipidemia)**
- Type III: Elevated IDL (remnant VLDL) **(Dysbetalipoproteinemia)**
- Type IV: Elevated VLDL (Pre-β lipoprotein) **(Familial prebetalipoproteinemia)**
- Type V: Elevated Chylomicrons and VLDL **(Familial combined hypertriglyceridemia)**

- Based on **fasting lipid profile** (Link to Frederickson phenotype)
  - **Hypertriglyceridemia**
    - **Type I**
    - **Type IV**
    - **Type V**
  - **Hypercholesterolemia**
    - **Type IIA**
  - **Hypertriglyceridemia and hypercholesterolemia**
    - **Type IIB**
    - **Type III**

- Recognize other names for each
- Identify molecular basis
- Clinical features
- Lab findings
- Differential diagnosis

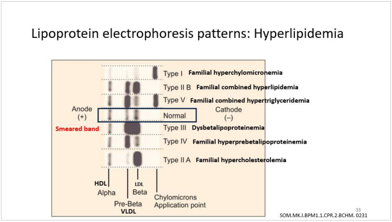
Familial **hyperchylomicron**emia: (Type I)

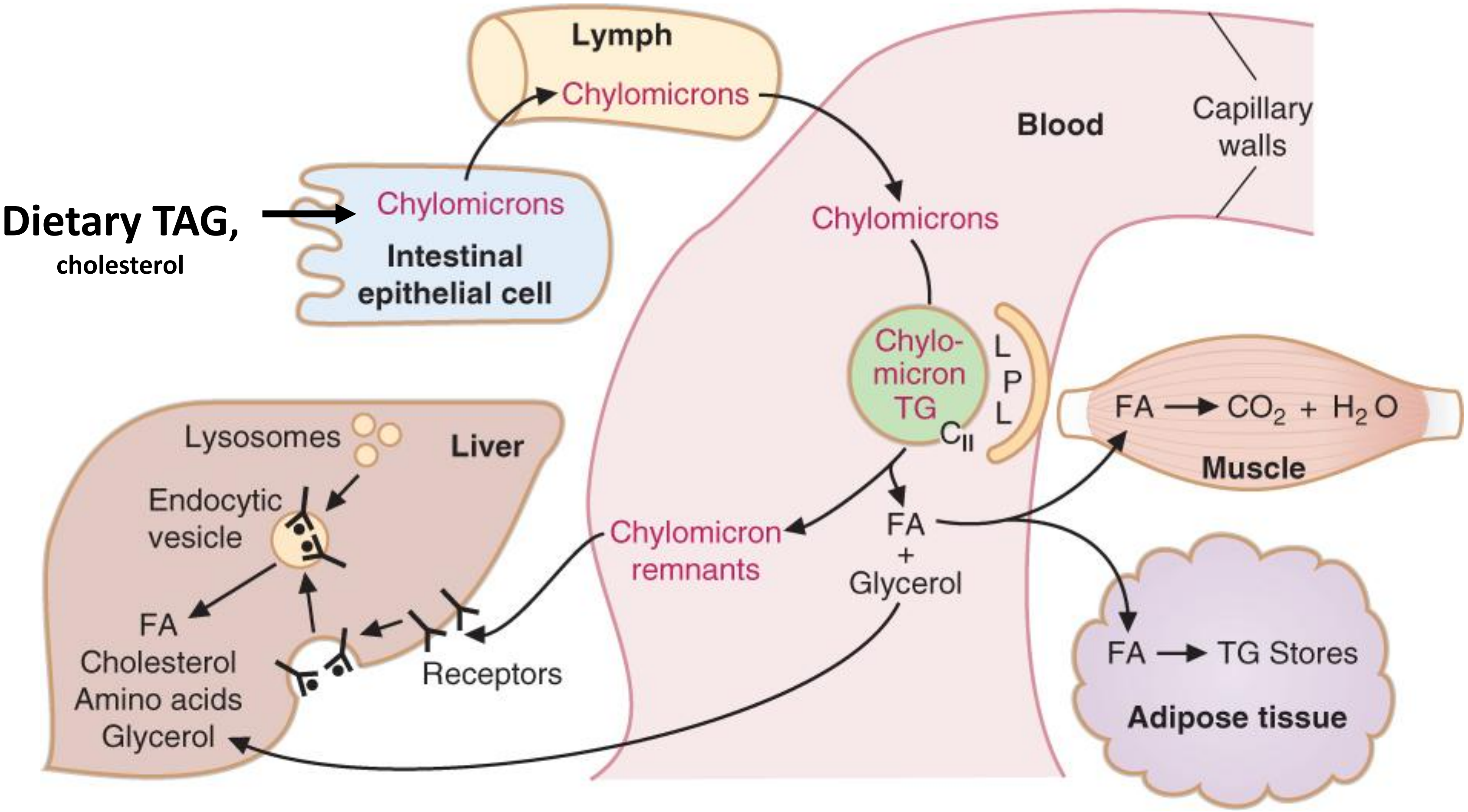
- Autosomal recessive
- Eruptive xanthomas: Trunk, extremities, buttocks
- Chylomicrons are large particles; Block capillaries in pancreas → **Acute abdominal pain** (**Recurrent pancreatitis**): Usually when TAG >500-600 mg/dL
  - Elevated serum amylase and lipase (pancreatitis)
- Plasma sample **turbid/ lipemic**
- **Creamy layer on top** on centrifugation (CMs least dense and float on surface)
- **Highly elevated serum triglycerides (TAG)**
- **Serum cholesterol: Normal**
- **Differentiate from types IV and V**



Familial hyperchylomicronemia: (Type I)

- Molecular defect: **Lipoprotein lipase deficiency**; Apo C-II deficiency
- **Lipoprotein electrophoresis**: Increased chylomicrons (Origin/ point of application)

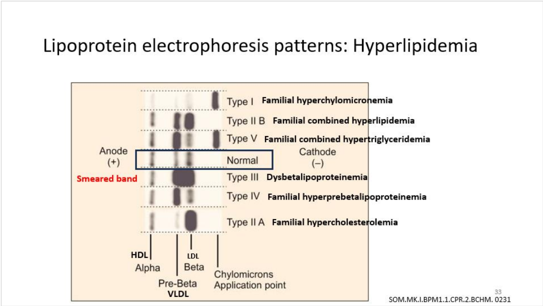




Familial **hyperprebetalipoproteinemia** (Type IV)

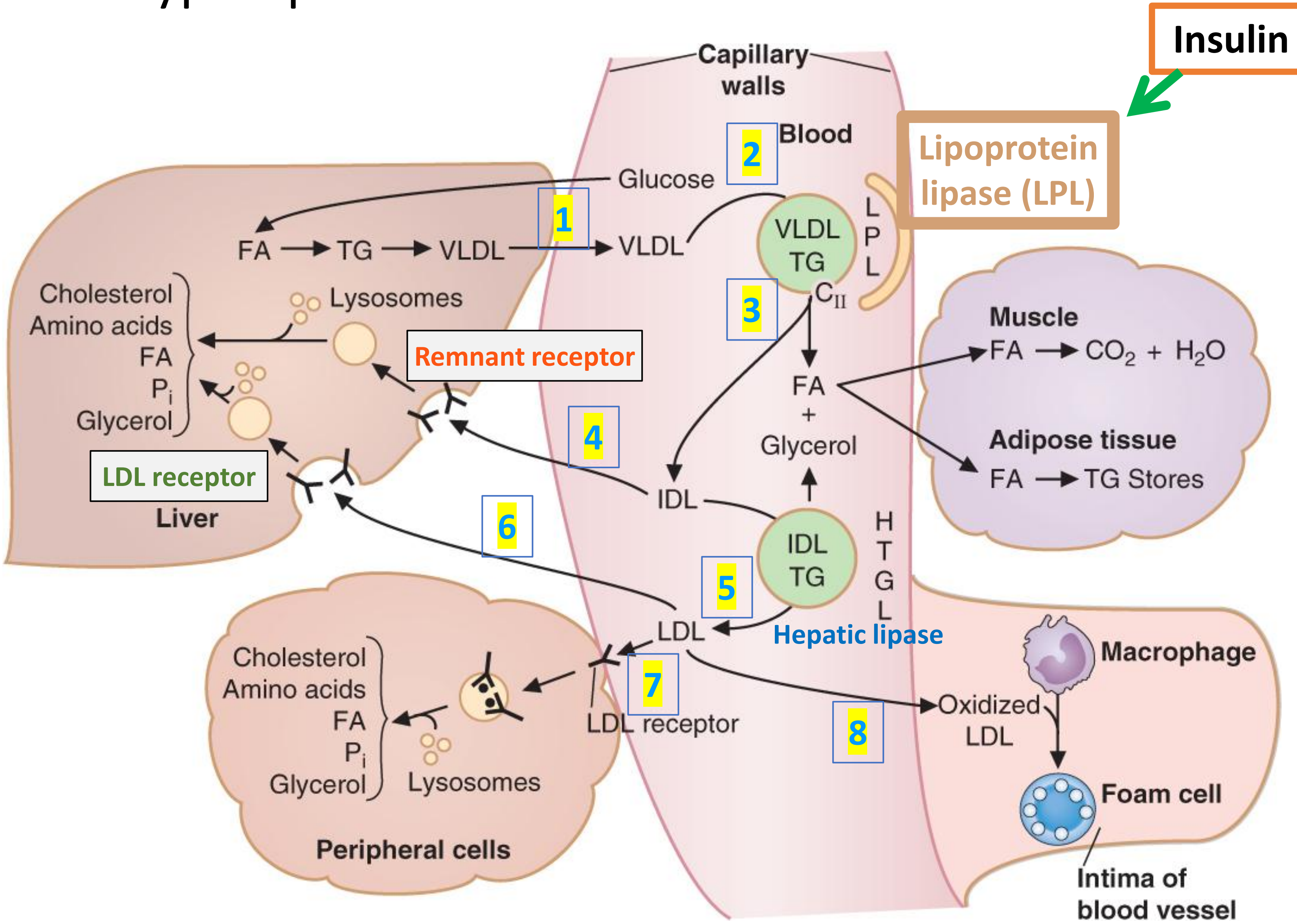
- Most common
- Eruptive xanthomas **(Very high triglycerides)**
- Plasma sample turbid/ lipemic
    - NO creamy layer on top after centrifugation
- **Elevated serum triglycerides (TAG)**
- Total serum cholesterol: Normal
- Low HDL-cholesterol levels (Usually associated)
- High TG/ HDL-cholesterol ratio
- Lipoprotein electrophoresis: Elevated VLDL (Pre-beta)
- Higher risk of **pancreatitis**
- Higher risk of **ASCVD** (Atherosclerotic cardiovascular disease)
- **Differentiate from types I and V**

Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition:* www.accessmedicine.com



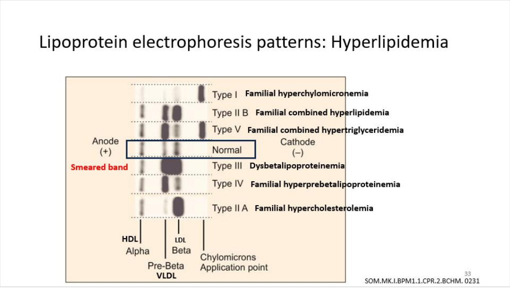
Molecular defects in Type IV hyperlipidemia

- Usually associated with obesity and type-2 diabetes mellitus
- Insulin resistance → Less activity of **Lipoprotein lipase (LPL)** → VLDL levels increase in blood
- **Apo C-II** deficiency → Low activity of LPL (3)



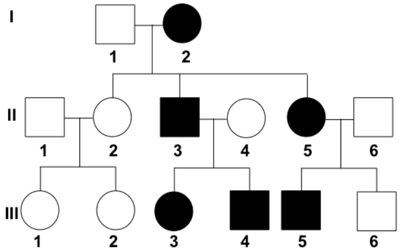
Familial combined hypertriglyceridemia (Type V)

- Plasma sample turbid
- Elevated serum triglycerides (TG)
- Serum total cholesterol: Normal
- Lipoprotein electrophoresis: **Elevated Chylomicrons and VLDL**

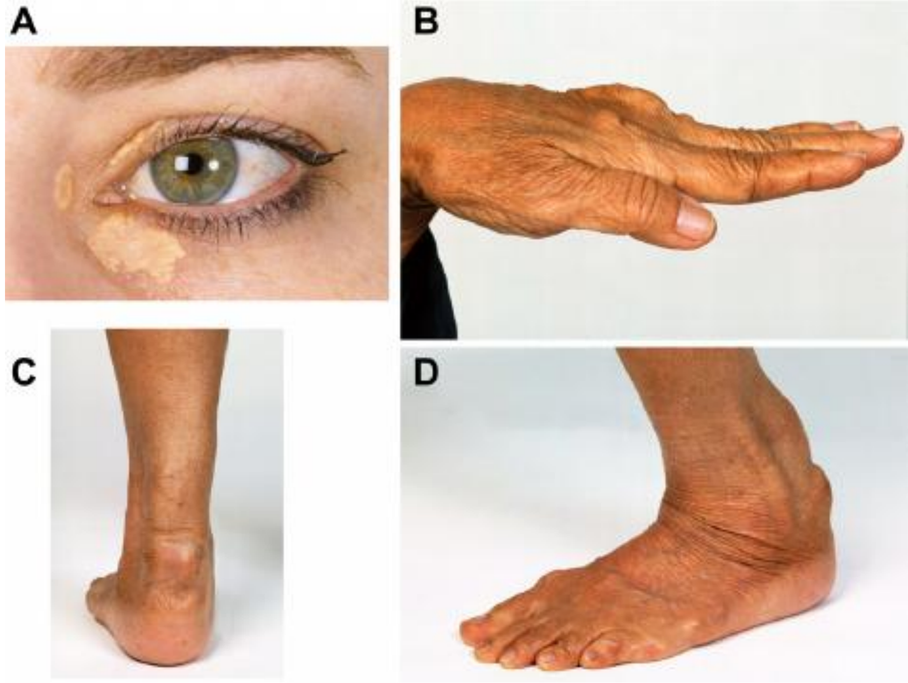


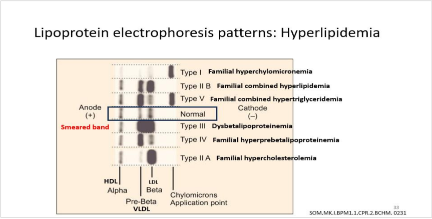
- Higher risk of **Acute pancreatitis**
- Eruptive xanthomas (Very high triglycerides)
- Higher risk of **ASCVD**
- Molecular defects:
  - Less activity of **Lipoprotein lipase (LPL)** → **VLDL and chylomicron** levels increase in blood
  - **Apo C-II** deficiency → Low activity of LPL (3) → VLDL and chylomicron levels increase in blood
- **Differentiate types I, IV and V: Lipoprotein electrophoresis**

Familial **hypercholesterol**emia: Type IIA



- **Autosomal dominant**; Young adults **(25-50 years)**
- Family history of early ASCVD/ hypercholesterolemia
- Tendon xanthomas (Achilles; Extensor surface of hand)
  - May be difficult to detect in early stages
- Xanthelasma (Lipid accumulation in eyelids)
- Corneal arcus (especially in young patients)
- Plasma sample clear (not lipemic/ turbid)
- **Elevated total cholesterol**
- **Elevated LDL-cholesterol**
- **Serum triglycerides (TAG): Normal**
- Lipoprotein electrophoresis:
  - Increased LDL (Beta-lipoprotein)





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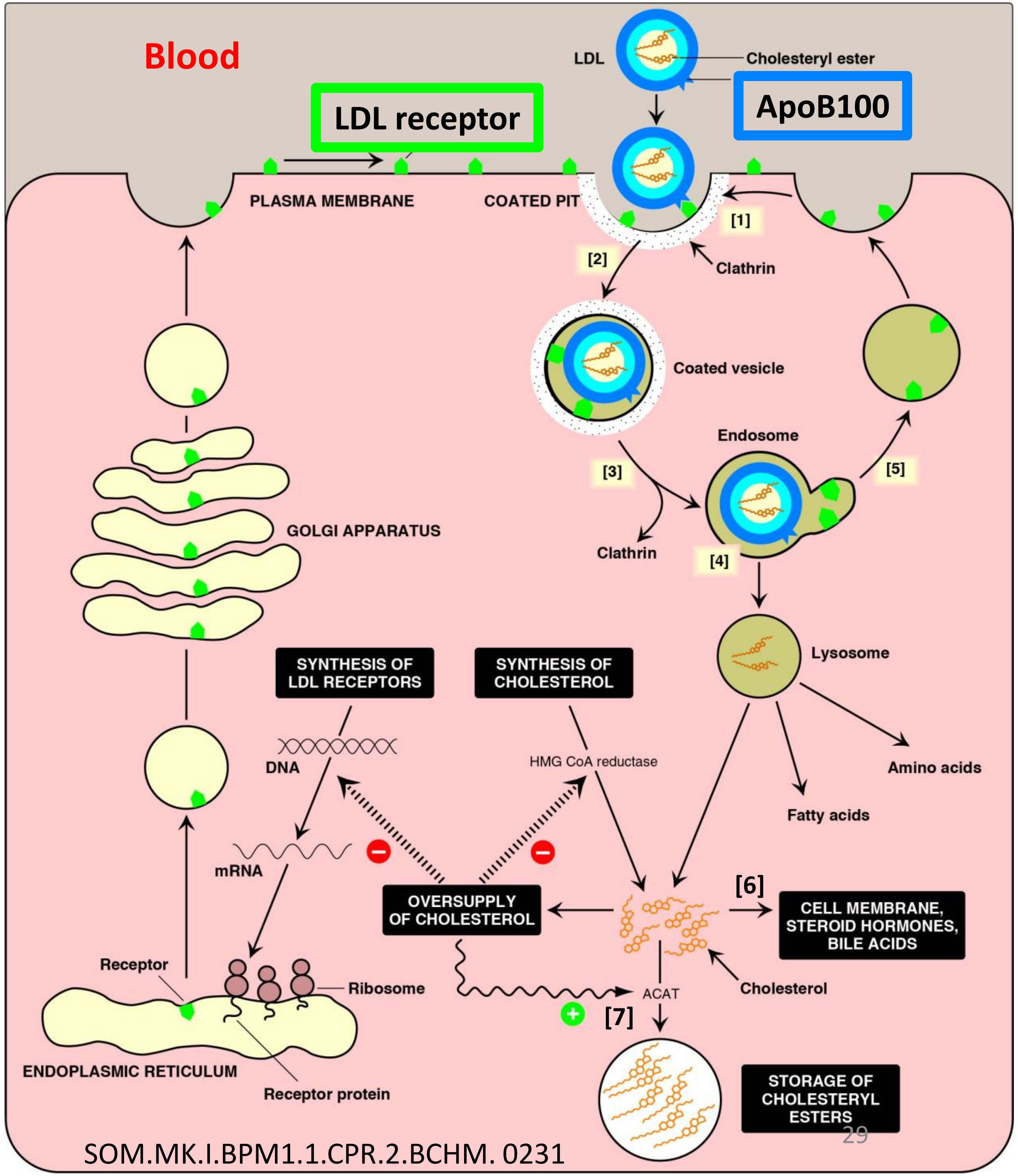
Familial hypercholesterolemia: Type IIA

1. **LDL receptors** present in plasma membrane of all cells (Clathrin coated pits)
2. LDL receptor binds to ApoB100 on LDL; **Receptor mediated endocytosis**

- **LDL receptor mutations → Defective receptor mediated endocytosis of LDL → Familial hypercholesterolemia**
- **B100 mutation (rare cause)**
- Most patients heterozygous (A*A) → 50% production of functional LDL receptor **(HAPLOINSUFFICIENCY); Allelic heterogeneity of LDLR mutations**
- Homozygous: Presentation in childhood; Highly elevated serum cholesterol and LDL-cholesterol levels
- When a patient diagnosed with Familial hypercholesterolemia, screen first degree relatives (parents, siblings and children)
- Management: Lifestyle modifications, dietary changes, statin, PCSK-9 inhibitors, Inclisiran to lower serum cholesterol



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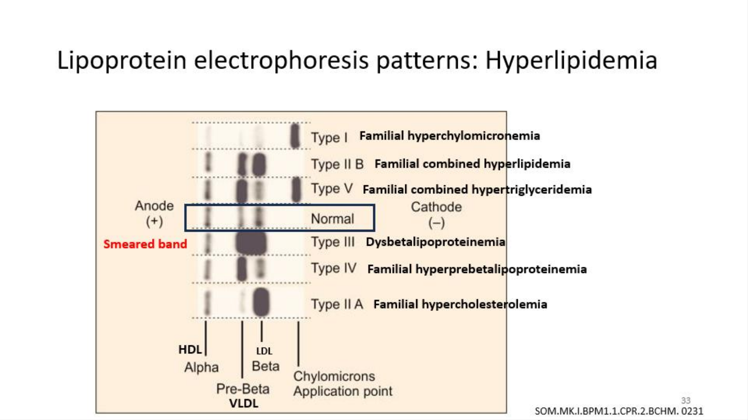
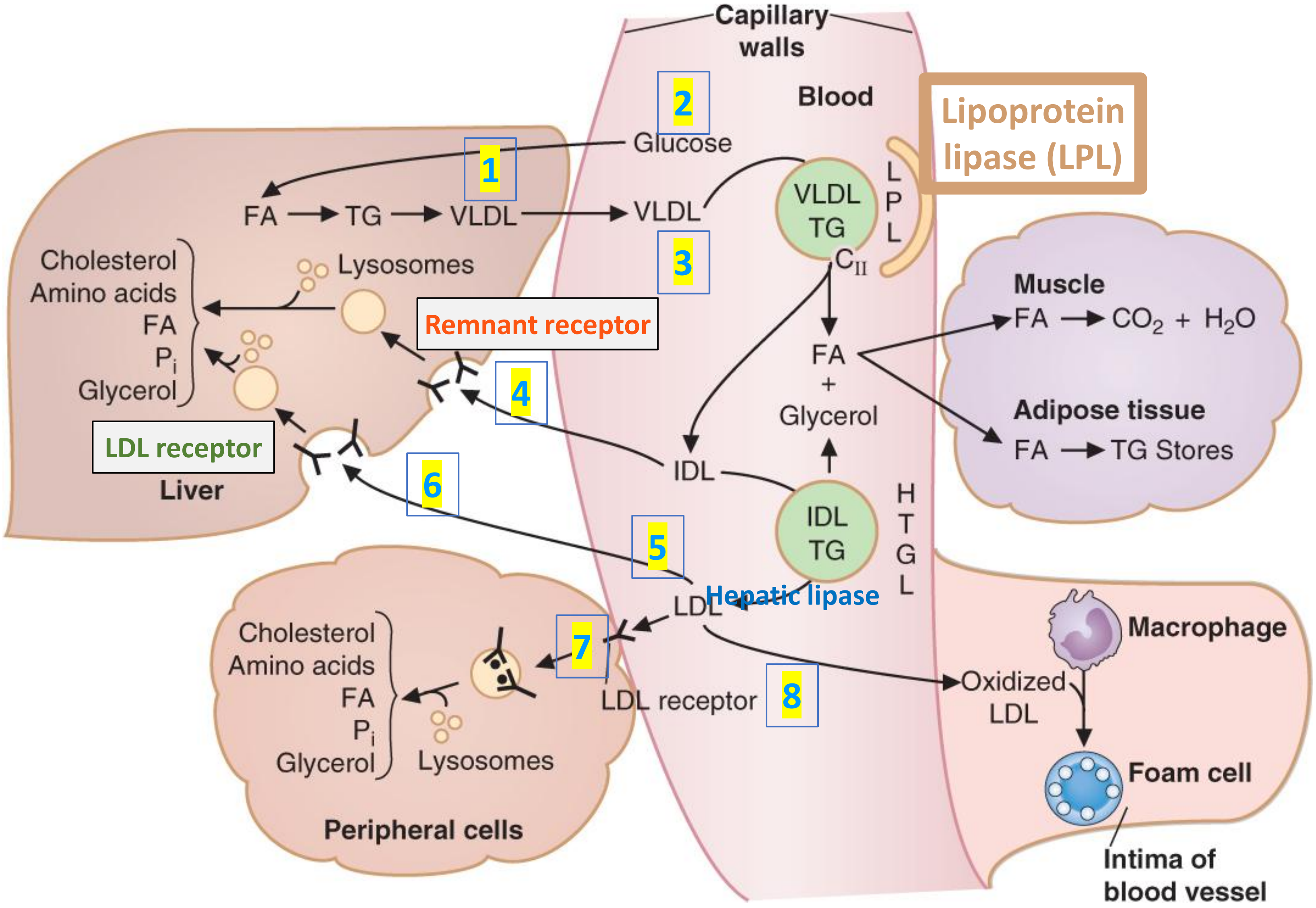
Dysbetalipoproteinemia (Type III)

- Age of presentation: 30-50 years
- **Palmar xanthomas**
- Plasma sample not turbid
- **Elevated serum triglycerides and serum cholesterol due to elevated IDL (remnant VLDL)**
- Lipoprotein electrophoresis: **Smeared band** between LDL and VLDL (Accumulation of IDL/remnant VLDL)
- **Differentiate from type IIB**



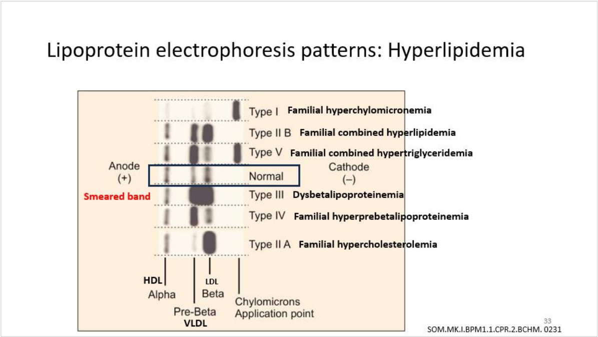
Molecular defect in dysbetalipoproteinemia (Type III)

- ==**Apo E mutations**== → defective recognition and uptake of IDL(4) (remnant VLDL)→ **Elevated IDL levels**
- Defective uptake of remnant chylomicrons

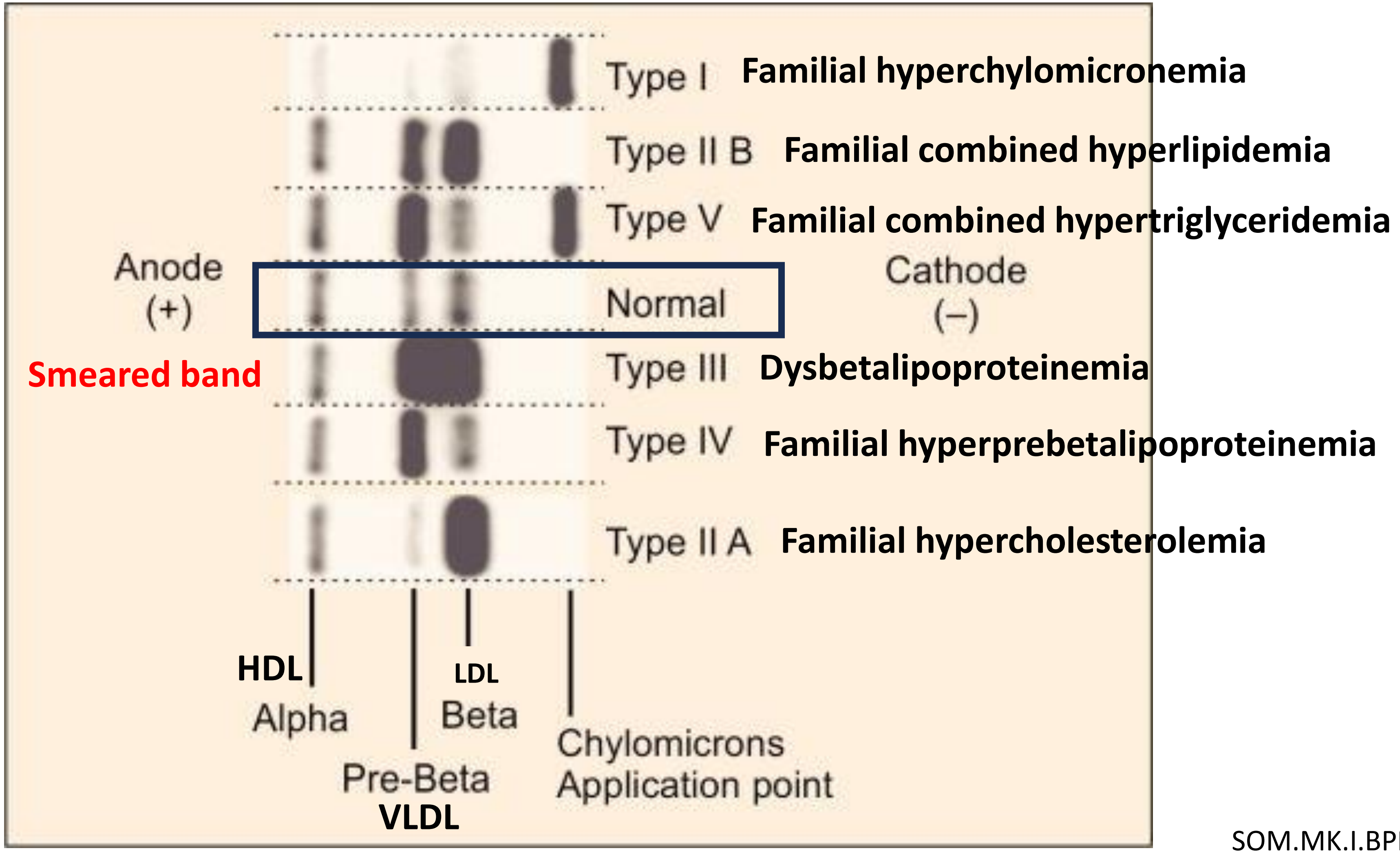


Familial **combined** hyperlipidemia (Type IIB)

- **Elevated serum total cholesterol, LDL-cholesterol and elevated serum triglycerides (TAG)**
- **Plasma** may be **turbid** due to elevated VLDL (No creamy layer on top)
- Lipoprotein electrophoresis: Elevated LDL (Beta) and VLDL (Pre-beta)
- **Xanthomas not common**
- Molecular defect: Usually multifactorial
  - Overproduction of VLDL
  - Defective LDL uptake/ metabolism
  - Higher LDL-B cholesterol
- More commonly associated with obesity, insulin resistance and type 2 diabetes mellitus



Lipoprotein electrophoresis patterns: Hyperlipidemia



Use the next few slides for self-study: Answers in practice questions

ENJOY THE LEARNING JOURNEY

Case report: 1

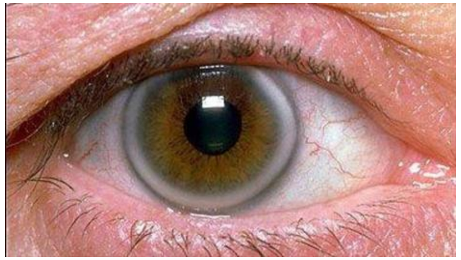
A 20-year-old man has the following lab reports:

	Patient	Reference range
Serum triglycerides (TAG)	140	35-160 mg/dL
Serum total cholesterol	450	<200 mg/dL
Serum HDL cholesterol	42	40-60 mg/dL

- Recognize other names for each
- Identify molecular basis
- Clinical features
- Lab findings
- Differential diagnosis

- Family history
- Physical findings
- Differential diagnosis

- Laboratory tests
  - Calculate LDL-cholesterol (Friedewald formula)



- Molecular basis:
- Management:

Case report: 2

A 26-year-old man comes to the physician for a routine examination.

	Patient	Reference range
Serum triglycerides (TAG)	540	35-160 mg/dL
Serum total cholesterol	190	<200 mg/dL
Serum HDL cholesterol	32	40-60 mg/dL

- Identify the possible hyperlipidemia phenotypes and their other names
- Physical findings and clinical features:
- Comment on appearance of plasma sample
- Identify the lipoproteins bands that are abnormal in the different hyperlipidemias (possible in this patient)
- Identify two molecular mechanisms for the hyperlipidemia
- Patient's BMI is 32kg/m^2 and he has central adiposity. Explain the lab findings?

A 30-year-old man has the following lab reports:

	Patient	Reference range
Serum triglycerides (TAG)	450	35-160 mg/dL
Serum total cholesterol	350	<200 mg/dL

- Recognize other names for each
- Identify molecular basis
- Clinical features
- Lab findings
- Differential diagnosis

- Physical findings
- Differential diagnosis
- Laboratory tests
  - Lipoprotein electrophoresis
- Molecular basis

A 45-year-old woman has the following lab reports:

	Patient	Reference range
Serum triglycerides (TAG)	300	35-160 mg/dL
Serum total cholesterol	250	<200 mg/dL
HDL cholesterol	35	50-70mg/dL

- Calculate LDL-cholesterol (Friedewald formula) and comment
- Lipoprotein electrophoresis
- Physical findings
- Risk factors
- Molecular basis
- Metabolic syndrome (insulin resistance syndrome)

A 5-year-old boy has the following lab reports:

	Patient	Reference range
Serum triglycerides (TAG)	100	35-160 mg/dL
Serum cholesterol	120	<200 mg/dL
HDL cholesterol	5	50-70mg/dL

- Physical findings
- Diagnosis
- Molecular basis

A 7-month-old boy has the following lab reports:

	Patient	Reference range
Serum triglycerides (TAG)	20	35-160 mg/dL
Serum cholesterol	40	<200 mg/dL

- Presenting findings
- Diagnosis
- Molecular basis

Hyperlipidemias: Summary

Disorder	Molecular defect	Clinical features	Laboratory findings
Type I (Familial hyperchylomicronemia)	**Lipoprotein lipase deficiency;** **ApoC-II deficiency**	**Eruptive xanthomas**, **Acute pancreatitis**; Higher risk of ASCVD	Elevated Triglycerides (TAG); Serum cholesterol: Normal; Plasma turbid with creamy layer on top; Elevated chylomicrons
Type IV (Familial prebetalipoproteinemia)			Elevated Triglycerides (TAG); Serum cholesterol: Normal; HDL-cholesterol: Low; Plasma turbid and **NO** creamy layer on top; Elevated VLDL
Type V (Familial combined hypertriglyceridemia)			Elevated Triglycerides (TAG); Serum cholesterol: Normal; Plasma turbid with creamy layer on top; Elevated chylomicrons and VLDL
Type IIA (Familial Hypercholesterolemia)	LDL receptor mutations → Defect in receptor mediated endocytosis	**Tendon xanthomas**, xanthelasma, corneal arcus; Higher risk of ASCVD (myocardial infarction)	Elevated total cholesterol and LDL-cholesterol; Triglycerides: Normal; Plasma NOT turbid; Elevated LDL
Type IIB (Familial combined hyperlipidemia)	VLDL overproduction; Defective LDL metabolism	Higher risk of ASCVD (myocardial infarction)	**Elevated TG, total cholesterol and LDL-cholesterol**; Plasma turbid (high VLDL); Elevated LDL (beta) and VLDL (pre-beta)
Type III (Dysbetalipoproteinemia)	ApoE deficiency	**Palmar xanthomas**; Higher risk of ASCVD	**Elevated TG, total cholesterol**; **Elevated IDL (remnant VLDL); Smeared band between LDL and VLDL**

Hypolipidemias: Summary

Disorder	Molecular defect	Clinical features	Laboratory findings
Abetalipoproteinemia	Microsomal TAG transfer protein deficiency	Fat malabsorption and steatorrhea (Fat in feces) Fat-soluble vitamin deficiency (A,D,E, K)	Fasting lipid profile: Low triglycerides; Low total cholesterol; Low LDL-cholesterol levels
Hypobetalipoproteinemia	Apo B mutation (Affects ApoB48 and ApoB100)		
Hypoalphalipoproteinemia (Tangier disease)	ABCA-1 transporter deficiency → defect in reverse cholesterol transport	Orange tonsils; Increased risk of ASCVD (Myocardial infarction)	Low HDL cholesterol

Functions of Apolipoproteins

Apolipo protein	**Lipoproteins in which present**	**Function**	**Deficiency features**
A-I	**HDL**	Activates LCAT and facilitates reverse cholesterol transport	Impairs LCAT activity → Low HDL-cholesterol→ Increase ASCVD risk
B48	**Chylomicrons and remnant chylomicrons**	Required for release from the intestinal mucosal cells into the lymph	Impairs release of chylomicrons into lymph→ Poor absorption of dietary lipids → Steatorrhea
B100	VLDL, IDL and **LDL**	VLDL: Required for release of VLDL from hepatocytes LDL: ApoB100 recognized by LDL receptor for receptor mediated endocytosis	Apo B100 mutations→ Defective recognition of LDL particles by LDL receptor (rare cause of familial hypercholesterolemia; Type IIA); Hypercholesterolemia
C-II	HDL, **Chylomicrons, VLDL**	HDL: Reservoir of ApoC-II for transfer to VLDL and chylomicrons VLDL and chylomicrons: ApoC-II activates lipoprotein lipase (LPL)	Impairs activity of Lipoprotein lipase → Elevated chylomicrons (Type I); Elevated VLDL (Type IV); or Elevated VLDL and Chylomicrons (Type V) → Increased TG; Pancreatitis; ASCVD
ApoE	HDL, VLDL, **IDL**, Chylomicrons and **remnant chylomicrons**	HDL: Reservoir of ApoC-II for transfer to nascent VLDL and chylomicrons IDL and CM remnant: Remnant receptor binds to ApoE in IDL and chylomicron remnants for receptor mediated endocytosis	Impaired uptake of IDL and chylomicron remnants by Remnant receptor → Dysbetalipoproteinemia (Type III); Elevated TG and total cholesterol;

DLA: Nutritional Aspects of Cardiovascular Disease

- Review after CPR lecture: HDL and lipoprotein disorders

SOM.MK.III.BPM1.1.CPR.2.BCHM. 0235	Discuss the major risk factors of cardiovascular disease.
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0236	Review dyslipidemia and lipoprotein patterns as a risk factor for coronary heart disease (CHD).
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0237	Highlight the significance of dietary fats related to CHD (Saturated/ Unsaturated; Omega-6 and omega-3 fatty acids; trans-fats)
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0238	Describe the utility of BMI and waist circumference to identify central obesity
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0239	Discuss the metabolic effects of obesity and the criteria for identification of metabolic syndrome (insulin resistance syndrome).
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0240	Discuss the metabolic effects of weight loss.
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0241	Review the role of dietary CHD risk factors, such as high salt and high glycemic index (refined sugars).
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0242	Highlight the role of protective factors against the development of coronary heart disease (CHD), such as fiber and vitamins that decrease homocysteine levels.
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0243	Review the role of antioxidants, vitamins and their long-term benefits.
SOM.MK.III.BPM1.1.CPR.2.BCHM. 0244	Describe benefits of DASH diet and Mediterranean diet.